

Adaptive Interaction Graphs

Research revision & evidence

October 5, 2026 | AISTATS revision | Local CPU and Metal experiments

The fixed study is complete; critique-driven follow-ups continue through the deadline.
General accuracy-runtime superiority remains unsupported.

The most important change is scientific: residual difficulty, the value of extra edges, and the cost of a changing rollout geometry must be measured separately. The revision repairs code, makes the graph budget explicit, and reports both favorable and unfavorable evidence.

What the audit establishes

Finding	Consequence
Likelihood mismatch	The released source uses $1/2$ where $d/2$ is required in the log-variance term and disagrees with the paper about whether Softplus predicts variance or standard deviation. Historical training provenance is incomplete.
Normalization mismatch	Released training and evaluation construct different scales. In WaterDrop the acceleration-scale ratios are 10.29 and 8.47. New full-model checkpoints store the trained normalization.
Original claim is too strong	WaterDrop MSE@200 improves by 0.327% in saved arrays; its conditional paired interval crosses zero. Full saved-horizon mean MSE is 3.86% worse than the sparse MSE baseline.
Edge savings are confounded	Expansion preserves base edges on identical positions. Lower edge counts on different predicted geometries cannot isolate efficient placement.

Deliverables

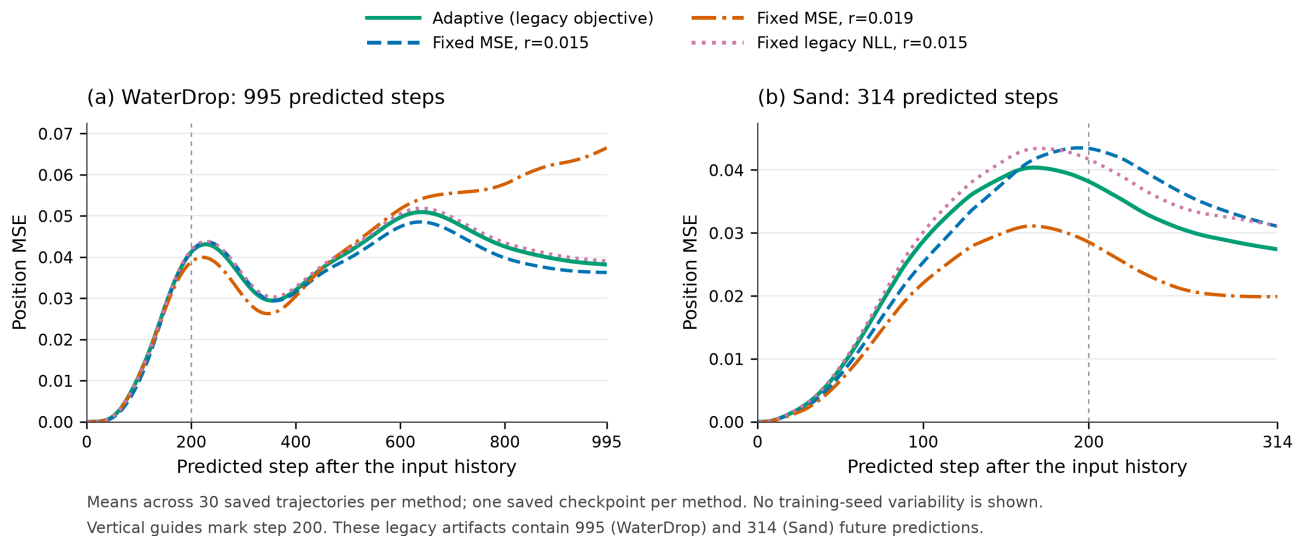
A compiling, editable manuscript with corrected definitions and proofs; tested research code; historical reanalysis; 15 compact models and six full-architecture models with 810 locked rollout outcomes; 1,782 saved-state action-benefit diagnostics; and a completed 2,550-frame graph-convention bridge. Restoring self-loops improves one-step accuracy but retains the risk-allocation deficit. The following pages distinguish each evidence source and its limits.

Source code: github.com/AidenZhou-Cohere/AdaptGNS-conference-revision
Branch: research/conference-revision. The manuscript remains a local working draft.

1. The original evidence, reanalyzed

Twenty committed artifacts were checked against their underlying per-trajectory arrays. Each historical method has one saved model; training-seed uncertainty cannot be reconstructed. The paired bootstrap assumes the same trajectory order because explicit IDs are absent. All intervals are exploratory and unadjusted.

Saved rollout errors across the complete prediction horizon



The plots use all 314 saved Sand forecasts and all 995 saved WaterDrop forecasts. Total recorded frames and future prediction steps are different; MSE at 1,000 future steps is unavailable.

WaterDrop metric	Adaptive	Sparse MSE	Difference / conditional 95% interval
MSE@200	.0412172	.0413525	-.0001353 [-.0008463, +.0005286]
MSE@500	.0411959	.0395592	+.0016367 [+.0000473, +.0031618]
Full-rollout mean	.0359455	.0346098	+.0013357 [+.0002413, +.0024355]
Mean edges	7,821.17	9,741.01	-19.71%; different predicted geometries

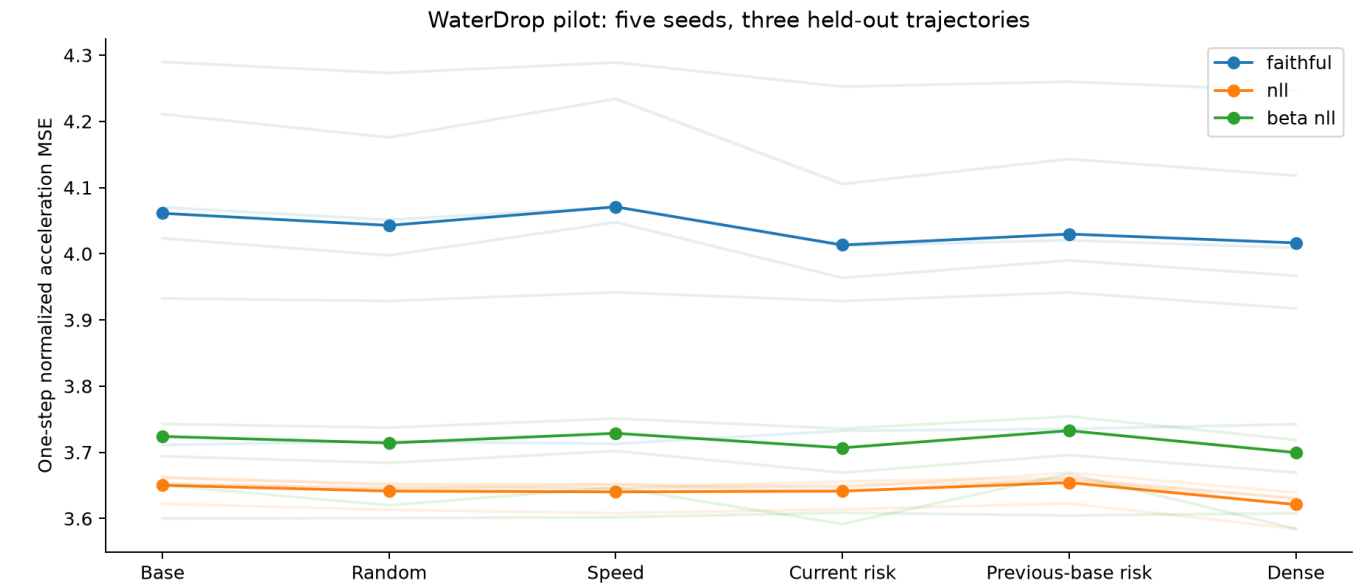
The adaptive policy still improves the reported NLL checkpoint in some comparisons. That is a useful narrower finding. It does not establish superiority over the stronger MSE alternatives. The saved hyperparameter sweep also lacks independent validation-selection provenance.

2. New controlled WaterDrop training

15 models completed 3,000 updates each: corrected NLL, beta-NLL (beta=0.5), and faithful regression, each with five seeds. The compact model has width 48, three message-passing blocks, six history frames, and no training noise. The convenience sample contains eight training trajectories and three each for validation and test.

The table reports normalized one-step acceleration MSE: mean +/- sample standard deviation over five seeds. Each seed equally averages 12 selected frames in each of three test trajectories. These are new pilot results, not the original benchmark or position rollout errors.

Objective	Base	Random	Speed	Current risk	Previous risk	Dense
faithful	4.061 +/- 0.223	4.043 +/- 0.212	4.071 +/- 0.225	4.013 +/- 0.192	4.030 +/- 0.196	4.016 +/- 0.187
nll	3.650 +/- 0.016	3.641 +/- 0.016	3.640 +/- 0.018	3.641 +/- 0.017	3.654 +/- 0.018	3.621 +/- 0.021
beta nll	3.724 +/- 0.128	3.714 +/- 0.132	3.729 +/- 0.132	3.707 +/- 0.136	3.733 +/- 0.129	3.700 +/- 0.133



Corrected NLL has the best mean one-step error here. Faithful regression is worse than the post-hoc zero-normalized-acceleration reference (3.80036), so this pilot does not establish it as an improved simulator.

All four allocation selectors use exactly one quarter of available annulus pairs on the same observed states. Current risk requires an extra scoring pass. Previous risk is evaluated on the previous observed base graph, not a free-rollout adaptive graph. These one-step comparisons isolate placement at a retained-edge budget, but do not establish long-horizon performance.

3. Residual risk versus useful intervention

The diagnostic freezes each trained model, predicts on a base graph and a dense graph at the same state, and keeps signed per-particle changes in squared residual. Positive benefit means the dense intervention reduced error. A high residual-risk correlation need not imply a high benefit correlation.

This distinction is mathematical as well as empirical. A perfectly calibrated score can assign risk 10 to unobserved forcing that extra neighbors cannot resolve, and risk 2 to a missing interaction whose inclusion reduces error by 1. Spending the budget on the highest-risk region is then the wrong action.

Objective	Risk-error correlation	Risk-benefit correlation
faithful	0.265 +/- 0.152	0.024 +/- 0.195
nll	0.419 +/- 0.074	-0.041 +/- 0.042
beta nll	0.336 +/- 0.052	-0.022 +/- 0.055

The full diagnostic records particle arrays and aggregates correlations within frames, then trajectories, then seeds. Dense-graph intervention benefit is not the marginal value of any single edge, and it is not a causal claim about the physical system. See risk_benefit.md for all objective-wise results and their provenance.

A budget with a precise meaning

All base pairs are retained. At most B optional undirected pairs are selected, and each is emitted in both directions. Thus the retained pair count is base + min(B, candidates). This bounds incremental message edges, not total memory or runtime: mandatory edges and candidate searches may still grow with density.

Under an additive edge-gain surrogate, top-B selection with score error at most delta loses at most 2 B delta. Actual set-benefit regret additionally needs uniform approximation error epsilon_B, giving 2 B delta + 2 epsilon_B. Perfect singleton scores alone can miss a complementary pair with arbitrarily large joint benefit. The appendix proves this conditional bridge and a mixed-difference bound; none is a measured simulator property or stability guarantee.

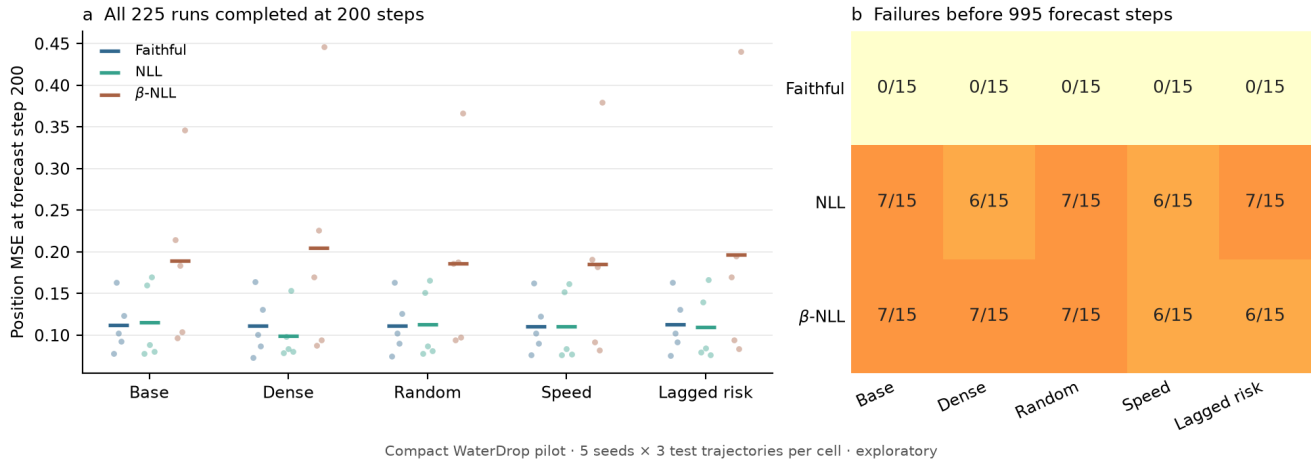
Measured CPU graph cost

Reference method	Median same-state time ratio to base
Dense radius	1.29x
Exact-budget random	1.38x
Exact-budget synthetic score	1.54x
70th-percentile synthetic score	1.67x

Nine observed WaterDrop states, five warmups, 40 randomized-order timing repetitions. Timings include SciPy tree construction, selection, and directed-edge materialization; exclude a neural network and learned scoring. The host was not isolated. These are CPU reference measurements and do not demonstrate a GPU or end-to-end speedup.

4. Autonomous rollout evidence

All 15 compact-model checkpoints were evaluated under five policies on three test trajectories. Only the initial six frames are observed; later histories, speeds and risk come from predicted states. This extension was added after inspecting the one-step pilot and is exploratory.



Left: dots are individual training-seed means over three trajectories; bars are five-seed means. Right: all failures are retained. Completing a numerical guard does not establish physical validity.

MSE@200	Base	Dense	Random	Speed	Lagged risk
faithful	0.1119	0.1112	0.1114	0.1108	0.1128
nll	0.1156	0.0990	0.1126	0.1103	0.1095
beta nll	0.1891	0.2048	0.1864	0.1854	0.1967

All 225 runs completed 200 steps. Risk allocation does not consistently improve error: NLL improves relative to its base policy, whereas faithful and beta-NLL worsen. Dense NLL is more accurate than its risk-budget policy at this horizon.

At 995 steps, 159 of 225 runs completed and 66 crossed the absolute-coordinate guard of 10. Additional guards cover nonfinite values and excessive candidate pairs. Full-horizon accuracy is undefined for any objective/policy group containing a failure; survivors are not substituted for the full group.

Only faithful groups completed all 15 trajectories. Their MSE@995 is 3.271 +/- 5.398 for base and 4.099 +/- 7.261 for lagged risk. Full-rollout mean MSE is 0.865 +/- 1.113 and 1.032 +/- 1.460, respectively. These large errors rule out interpreting guard completion as stable, accurate simulation.

The two horizons reproduce every 200-step error/count prefix exactly. Cached risk forecasts first on base, then applies the quarter-annulus budget; random and speed apply it immediately. Autonomous geometries differ, so total counts are not matched. Timings include expanded candidate search and the risk head even for base. An optimized mean-only GNS comparison remains unmeasured. See pilot_rollout_results.md for outcomes and hashes.

5. Candidate improvement and larger experiments

A training-only benefit head

A fixed ridge predictor was fitted on detached embeddings and signed dense-intervention labels from the 192 training frames of each frozen model. No validation/test labels entered fitting. This is a post-hoc candidate motivated by the pilot, with no independent confirmation or marginal-edge-value claim.

Test one-step MSE	Random25	Benefit25	Benefit-label correlation
Faithful	4.0428	4.0529	0.060 +/- 0.075
NLL	3.6414	3.6319	0.005 +/- 0.032
Beta-NLL	3.7141	3.7053	0.023 +/- 0.044

NLL and beta-NLL beat random in all five seeds, but faithful worsens. Dense allocation has lower mean error than the benefit-head policy for each objective. Weak label correlations and the additional scoring pass limit what these improvements establish.

The full model can use local Metal

The original 1.625M-parameter simulator runs natively on the Apple GPU with CPU fallback disabled. Measured graph/transfer-inclusive Adam steps were 68.85 ms for 678 particles and 90.00 ms for 1,033 particles. Arithmetic 500,000-update projections are 9.56–12.50 hours, excluding full-loader, validation, checkpoint and sustained-load overhead.

Forward parity is close, but some strict gradient comparisons fail (one recorded relative L2 difference is 0.0825%). The device report retains this failure and analyzes ReLU branch sensitivity; these measurements do not certify identical optimization trajectories. A production Metal path and larger data experiment are tracked separately.

Implementation and next decisions

The original architecture passed a real train/reload/validation/short-rollout smoke pipeline. Contract tests cover likelihood dimensions, faithful mean gradients, checkpoint semantics, split integrity, exact budgets, calibration, and rollout information flow. These checks establish execution, not scientific performance.

All six local Metal models have completed 100,000 updates: all official training data, the original architecture, noise augmentation, two objectives and three paired seeds. Their final hashes and validation records passed the completion gate. The locked test evaluation completed October 5 at 20:22 UTC; all eight scientific guard failures are retained. This budget remains shorter than the historical 500,000-update benchmark. See `full_experiment_protocol.md` for failure handling and the decision rule.

The author must verify the manuscript, proofs, citations, results and AI Use Statement. The abstract is registered. Full paper and supplement are due October 7, 2026 at 7:59 a.m. Toronto time (October 6, 23:59 AoE). Native compilation passes an eight-page main-text assertion; final PDF visual review remains outstanding.

Official deadline: virtual.aistats.org/Conferences/2027/CallForPapers. This report does not certify conference readiness.

6. Six full models: paired validation

All six models completed the fixed 100,000 updates on all 1,000 training trajectories with the original width-128, ten-block architecture. All final hashes are verified. The locked test evaluation is complete; this section concerns validation only.

Snapshot: 2026-10-05T16:03:27.535178+00:00. Recorded updates: 600,000 / 600,000.

Clean validation, not policy or rollout results

Final model / seed	Coordinate MSE	Binned vector-risk gap	Gaussian NLL
faithful / 0	0.00811862	0.01228642	-4.28672
nll / 0	0.00945431	0.00663060	-4.48935
faithful / 1	0.00905942	0.00831614	-4.30024
nll / 1	0.01003003	0.00484836	-4.43652
faithful / 2	0.00875011	0.01110184	-4.22028
nll / 2	0.00897528	0.00754457	-4.54691

Each endpoint averages 128 fixed clean histories within 30 validation trajectories, then trajectories equally, using stored noise-adjusted normalization. These are not compact-pilot units. All 126 scheduled records remain in `full_validation_curves.json`.

Faithful final mean +/- sample SD across three seeds: MSE 0.00864272 +/- 0.00047951; gap 0.01056813 +/- 0.00203824; Gaussian NLL -4.26908 +/- 0.04280.

NLL final mean +/- sample SD: MSE 0.00948654 +/- 0.00052811; gap 0.00634118 +/- 0.00137120; Gaussian NLL -4.49093 +/- 0.05521.

NLL versus faithful (seed 0: 16.45% higher MSE, 46.03% lower gap; seed 1: 10.71% higher MSE, 41.70% lower gap; seed 2: 2.57% higher MSE, 32.04% lower gap) retains the error--risk-scale tradeoff in all three pairs; Gaussian NLL is also lower in each.

Paired NLL-minus-faithful differences (mean +/- sample seed SD): MSE: +0.00084382 +/- 0.00056601; gap: -0.00422696 +/- 0.00123824; NLL: -0.22185127 +/- 0.09662048. Within-seed percent changes average +9.91 +/- 6.97% in MSE and -39.92 +/- 7.16% in gap; these are not ratios of group means or significance claims.

Final versus 95k: faithful 0: MSE -7.73%, gap +72.33%, NLL worsens; nll 0: MSE -0.0053%, gap +12.37%, NLL worsens; faithful 1: MSE +2.74%, gap -23.51%, NLL improves; nll 1: MSE +6.62%, gap -43.76%, NLL improves; faithful 2: MSE -5.32%, gap +16.21%, NLL improves; nll 2: MSE -5.15%, gap +16.27%, NLL improves. Earlier reversals remain in the saved arrays. Final checkpoints stay fixed. These diagnostics establish neither convergence, calibration, useful allocation, test performance nor a speed advantage.

Completion cost and locked evaluation

Faithful seed 0 finished October 4 at 21:58:57 UTC. Its original-start-to-finish span was 20,485.286 seconds (5 h 41 m 25 s), including restart and downtime; retained lineage records 19,550.767 seconds. Rollback omitted at least 588.212 timer seconds and replayed at least 1,600 updates. Attempt records account for at least 20,138.979 seconds. These are separate timestamp and elapsed-counter measures, not GPU compute time.

October 5 UTC finish / elapsed timer: nll 0: 01:36:20 / 13,023.416 s; faithful 1: 05:09:52 / 12,794.132 s; nll 1: 08:32:09 / 12,124.411 s; faithful 2: 11:46:32 / 11,642.209 s; nll 2: 15:58:10 / 13,306.718 s. NLL seed 0 started October 4 at 21:59:07 UTC without a recorded restart. The same supervisor completed all six jobs. Workload differences prevent a fair speed comparison.

The faithful seed 0 restart restored the verified 20k model, Adam and RNG state, but Metal losses were not bitwise identical. All interrupted-attempt records remain.

NLL seed 2 timestamps span 15,085.684 seconds, while its retained elapsed counter is 13,306.718 seconds. Two monitored intervals separately differed by 1,080.922 and 689.911 seconds. Their cause remains unverified: these are not established downtime, sleep or GPU-utilization measurements. Previous projections and all clock records remain preserved.

Six retained trainer counters sum to 82,441.655 seconds (22.900 hours); current job timestamp intervals sum to 75,750.066 seconds (21.042 hours). The resumed-first-job-to-last-completion timestamp envelope is 75,806.731 seconds. The first job starts after recovery while its counter retains earlier work; these totals do not measure the same interval or exact supervisor lifetime.

Evaluation completion

At 16:05 UTC on October 5, the completion and process-inactivity gate passed. The official test source matched its pinned SHA256, and conversion verified all 30 ordered records. The native MPS queue completed indices 3-29 under all five locked policies at 20:22 UTC. All autonomous and same-state placement, signed-benefit and repeated-runtime jobs are complete. Cutoff: October 7, 08:00 UTC.

Preserve guard-failed trajectories as outcomes. All-sample full-horizon accuracy remains undefined for groups containing failures; survivor means cannot replace it. Report physical boundary excursions even when numerical guards pass. Historical aggregates and the first three test trajectories were already inspected; this is not pristine independent confirmation.

The locked 27-trajectory, five-policy rollouts and same-state diagnostics are complete; pages 11-13 report their outcomes. Noisy-to-clean risk calibration also need not transfer; the separate derivation retains the conditional residual-noise cross term.

7. Risk and observed physical descriptors

A separate exploratory CPU postprocess reuses saved scores from all 15 compact models: 36 validation and 36 previously inspected test frames per model. No new model inference or training was required. These associations are not independent confirmation and do not change the fixed full-model experiment.

Local strain and rotation

Using only positions before the prediction target, fit local velocity differences to position differences within radius 0.015. The gradient requires at least three neighbors, full spatial rank and Gram condition at most one million. Strain is the Frobenius norm of its symmetric part; vorticity is the absolute difference of its off-diagonal entries. These are observed kinematic descriptors, not ground-truth physical complexity.

Within each frame, compute average-rank Spearman correlation with saved base-graph risk. The adjusted variant regresses both rank vectors on ranked neighbor count, speed and signed wall clearance. Average equally over frames, three trajectories and five seeds. The table gives mean +/- sample seed SD, not confidence intervals or particle-level p-values.

Objective	Split	Strain	Adjusted strain	Vorticity	Adjusted vorticity
faithful	valid	0.312 +/- 0.041	0.160 +/- 0.027	0.181 +/- 0.032	0.084 +/- 0.026
faithful	test	0.236 +/- 0.021	0.167 +/- 0.025	0.098 +/- 0.009	0.041 +/- 0.019
nll	valid	0.442 +/- 0.037	0.311 +/- 0.033	0.212 +/- 0.036	0.077 +/- 0.024
nll	test	0.397 +/- 0.057	0.283 +/- 0.059	0.213 +/- 0.052	0.061 +/- 0.053
beta nll	valid	0.371 +/- 0.020	0.232 +/- 0.028	0.191 +/- 0.022	0.067 +/- 0.020
beta nll	test	0.310 +/- 0.011	0.196 +/- 0.014	0.157 +/- 0.015	0.026 +/- 0.016

Strain retains positive mean association after adjustment. Adjusted vorticity is weaker. The shared geometry mask covers 95.99% of validation and 96.74% of test particle-frame pairs; coverage varies from 88.05% to 99.66% across test frames. Every required frame coefficient is defined. Excluded particles and their score distributions are retained.

Spatial alignment does not establish useful allocation

Test objective	Risk-error, same mask	Risk-dense-benefit, same mask
faithful	0.254 +/- 0.150	0.025 +/- 0.193
nll	0.411 +/- 0.075	-0.051 +/- 0.045
beta nll	0.327 +/- 0.052	-0.042 +/- 0.058

Whole-dense benefit shares base error algebraically and is neither necessary nor sufficient for useful selected-edge allocation. Actual budgeted-policy accuracy, failures and runtime remain the allocation tests. The next page reports a separate exploratory comparison of two cheap physical allocation scores on the compact models.

The complete physical_complexity_pilot.md/.json retain both splits, all seeds, all secondary descriptors, signs and coverage. Source/protocol/input hashes were fixed before these associations were computed. The postprocess took 2.943 CPU seconds, which is not policy latency. All 122 synthetic tests and 33,070 independent cached-array checks passed.

8. Cheap physical allocation controls

A separate exploratory follow-up evaluates two observed-state heuristics on all 15 frozen compact models. It uses the same 36 frames in each previously inspected split, every particle and the original float32 graph convention. Both scores keep all mandatory base pairs and exactly one quarter (rounded down) of the available annulus pairs.

Inverse count ranks nodes by negative base degree. Velocity RMS ranks them by the root mean squared difference between their last observed displacement and their base neighbors; isolated nodes score zero. Pair priority is the maximum endpoint score, with the original ID tie rule. Neither score uses a prediction target.

Paired one-step test error differences

Percent difference is computed within each seed, then averaged across five seeds; +/- is sample seed SD. Negative favors the physical policy. Each seed gives equal weight to 12 frames per trajectory and three trajectories. All six replayed controls, both splits and all raw outcomes are retained in physical_allocation_pilot.md/.json.

Objective	Physical score	vs random	vs current risk	vs previous risk
faithful	Inverse count	+0.429% +/- 0.463%	+1.154% +/- 1.167%	+0.740% +/- 0.839%
faithful	Velocity RMS	-0.011% +/- 0.238%	+0.710% +/- 0.987%	+0.299% +/- 0.713%
nll	Inverse count	-0.002% +/- 0.060%	-0.001% +/- 0.180%	-0.353% +/- 0.041%
nll	Velocity RMS	-0.309% +/- 0.181%	-0.309% +/- 0.177%	-0.659% +/- 0.219%
beta nll	Inverse count	+0.176% +/- 0.170%	+0.380% +/- 0.557%	-0.325% +/- 0.295%
beta nll	Velocity RMS	-0.432% +/- 0.506%	-0.231% +/- 0.201%	-0.928% +/- 0.929%

NLL velocity RMS improves over random, speed, current risk and previous risk in all five test seeds; dense remains better on average. Faithful test error instead favors the risk policies. For beta-NLL, the mean within-seed difference from random reverses sign: -0.432% on test but +0.883% on validation. There is no general winner.

Replay, ties and limits

All 6,480 original control replays passed the fixed MSE tolerance and exact edge counts. The independent cached-array audit passed 84,612 checks, including all 1,080 raw-frame hashes and a predetermined 90-frame numerical sample. All 54 focused synthetic tests passed before inference. Source, protocol and inputs were fixed before this attempt.

The budget splits score ties on 35/36 validation and 36/36 test frames for inverse count, and 27/36 and 21/36 for velocity RMS. The deterministic ID rule therefore matters to selection; prediction sensitivity to relabeling was not measured. There were 37/53 isolated particle-frame pairs in validation/test, all retained.

The complete single-thread CPU attempt took 37.737 seconds with shared work and concurrent full-model training. This is an operational measurement, not a fair policy latency comparison. These one-step results are exploratory after prior test inspection; they establish neither autonomous rollout stability nor full-architecture superiority. The locked five-policy full-model evaluation is unchanged.

10. Full architecture: locked rollout outcomes

All six final 100,000-update models were evaluated on official test indices 3-29 for 995 forecasts under all five declared policies. The queue completed at 20:22 UTC on October 5. Prior inspection of indices 0-2 and historical aggregates prevents a pristine independent-confirmation claim.

All 405 faithful outcomes completed. NLL has eight coordinate-guard failures, all in seed 2: base 1, dense 2, random 4 and cached risk 1. Speed has no guard failures. Failed outcomes are preserved, and affected all-sample full-horizon means remain undefined.

Objective / policy	Mean MSE	MSE at 200	Failures / 81
faithful / Base	0.02665 +/- 0.02015	0.01289 +/- 0.00858	0
faithful / Dense	0.03823 +/- 0.01880	0.02234 +/- 0.02031	0
faithful / Random25	0.02737 +/- 0.02010	0.01459 +/- 0.01243	0
faithful / Speed25	0.02519 +/- 0.01477	0.01278 +/- 0.00777	0
faithful / Cached risk25	0.02820 +/- 0.01786	0.01402 +/- 0.01094	0
nll / Base	undefined	0.02770 +/- 0.02460	1
nll / Dense	undefined	0.02712 +/- 0.02222	2
nll / Random25	undefined	0.02663 +/- 0.02360	4
nll / Speed25	0.03210 +/- 0.01408	0.02726 +/- 0.02314	0
nll / Cached risk25	undefined	0.02528 +/- 0.02419	1

Errors average coordinates, particles and 995 forecasts within each trajectory, then 27 trajectories within a seed. +/- is sample SD across exactly three training-seed means. MSE at 200 remains defined because all failures occurred later. All final endpoints and per-seed values are retained in `full_rollouts.json`.

Faithful cached-risk-minus-speed mean-MSE difference is +0.003008 +/- 0.003151, positive in every seed. Differences from base (+0.001555 +/- 0.002630) and random (+0.000827 +/- 0.002250) change sign across seeds. Cached risk beats dense in every faithful seed. Three seeds support descriptive comparisons, not strong significance claims.

11. Completion is not physical validity

All eight NLL failures occurred during a forward attempt when the maximum absolute coordinate exceeded the declared guard of 10. Accepted prefixes range from 876 to 918 forecasts; the rejected attempt is excluded. No guard failure was retried.

Policy	Source index	Accepted forecasts	Outcome
Base	17	918	coordinate guard
Dense	16	901	coordinate guard
Dense	17	912	coordinate guard
Random25	6	898	coordinate guard
Random25	20	918	coordinate guard
Random25	26	876	coordinate guard
Random25	29	893	coordinate guard
Cached risk25	29	891	coordinate guard

The geometric diagnostic compares forecasts with metadata bounds using a 1e-6 tolerance. It averages the fraction of particles outside over steps, then trajectories and seeds. These are not calibrated physical failure thresholds.

Complete group	Outside particles (%)	Mean trajectory max excursion
faithful / Base	13.46 +/- 6.00	0.2393 +/- 0.1082
faithful / Dense	14.88 +/- 5.29	0.3868 +/- 0.0807
faithful / Random25	13.61 +/- 6.25	0.2681 +/- 0.0964
faithful / Speed25	13.22 +/- 6.00	0.2728 +/- 0.0852
faithful / Cached risk25	13.11 +/- 6.10	0.3016 +/- 0.1120
nll / Speed25	11.74 +/- 7.77	0.5650 +/- 0.3674
Same-frame ground truth	2.6777	0.007820

The final column averages each trajectory's worst excursion; it is not the single worst particle across the study. Failed NLL groups have undefined full-horizon boundary summaries, including their truth references. Their accepted-prefix diagnostics remain separately available; no survivor-only summary replaces them.

Even the completed faithful trajectories leave the metadata bounds much more often and farther than ground truth. Neither absence of a numerical guard failure nor improved one-step error certifies plausible fluid dynamics. Energy, momentum and other conservation diagnostics remain unmeasured.

12. Common states: allocation and cost

All six models completed 297 fixed observed histories: 27 trajectories times 11 target frames. Budgeted policies retain exactly one quarter (rounded down) of annulus pairs and preserve the common base graph. Previous observed-base risk requires a separate scoring pass; it is distinct from autonomous cached risk.

Policy	Faithful MSE	NLL MSE	Faithful / NLL ms
Base	0.00959 +/- 0.00014	0.01061 +/- 0.00051	10.74 / 10.50
Dense	0.01192 +/- 0.00085	0.01237 +/- 0.00019	13.86 / 13.59
Random25	0.00899 +/- 0.00026	0.00984 +/- 0.00014	11.80 / 11.55
Speed25	0.00943 +/- 0.00014	0.01058 +/- 0.00020	11.78 / 11.52
Previous observed risk25	0.00947 +/- 0.00026	0.01070 +/- 0.00007	22.38 / 21.88

MSE is normalized acceleration coordinate error, not autonomous position error. Timings are milliseconds, six synchronized rotated repeats after warmup per case/frame, then equal-frame, equal-trajectory and three-seed averages. All risk-head work, graphs, selection, transfers, network and decoding are included; residual analysis and file writing are excluded.

Previous-risk Spearman	Faithful	NLL
With base error	0.357 +/- 0.060	0.411 +/- 0.023
With signed dense benefit	-0.022 +/- 0.045	-0.048 +/- 0.012

Mean risk correlations with signed dense benefit are weakly negative, despite positive correlations with residual error. Dense raises average one-step error for both objectives; random25 beats previous risk in every seed of both objectives. Mean signed normalized-vector benefits are -0.004649 +/- 0.001893 (faithful) and -0.003511 +/- 0.000642 (NLL). Dense benefit is not marginal single-edge value.

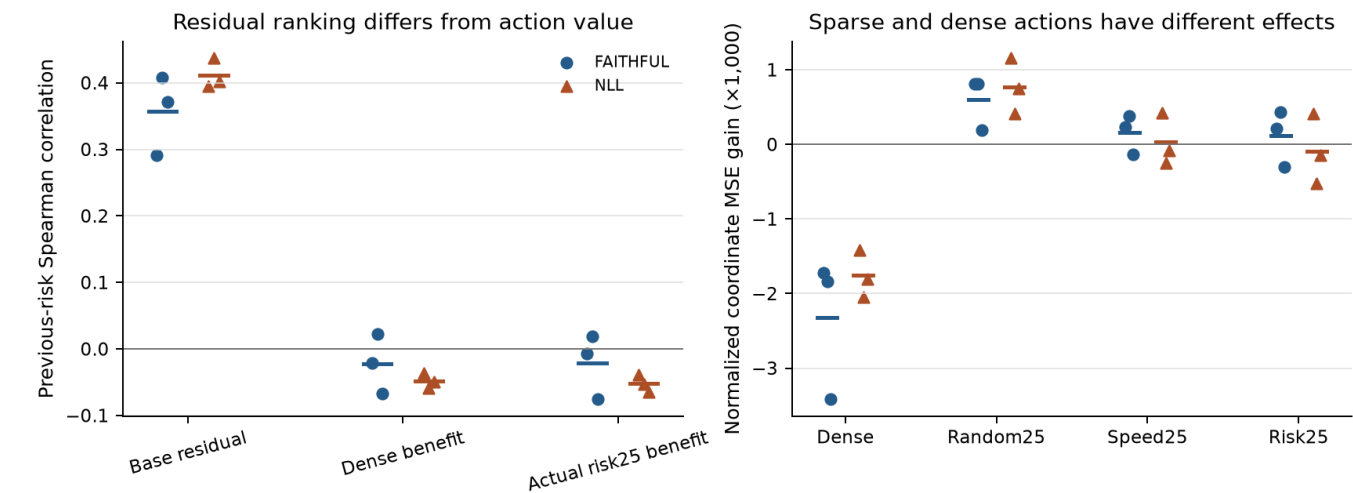
Natural base costs 10.740/10.496 ms; imposing the expanded candidate search costs 11.067/10.815 ms. The extra previous-risk scoring pass costs 10.596/10.347 ms. Every policy runs the risk head. Peak memory and an optimized mean-only baseline are unmeasured; the shared host was not isolated. Autonomous timings have fixed policy order, different geometries and some shorter failed horizons, so they cannot establish speedup.

The two radius classifiers produced identical pairs on all 297 observed geometries. One retained NLL seed-2 case (source 6, target 106) produced two risk-selected pair hashes across six timed calls; the maximum recorded prediction difference was 2.45e-5. Exact budgets remained satisfied, and no repeat was discarded or rerun.

Decision: retain the diagnostic allocation framing. The complete study does not support general cached-risk accuracy-runtime superiority. Author verification, registered-abstract comparison, final PDF inspection, venue eligibility and anonymous supplement/asset-rights checks remain required.

13. Actual actions, abstention and the next controls

After inspection of the fixed study, we reanalyzed all 1,782 observed-history cases, each with five saved policy predictions. This is a separate exploratory analysis with no new training or inference. Every original and derived file passed independent numerical and provenance audit.



Each symbol is an equal-trajectory seed mean; short bars are means of three seeds. Positive MSE gain favors the action. Risk25 uses previous observed-base risk; these are teacher-forced one-step results under the original no-self-loop evaluation convention. Dense has a larger edge budget.

Diagnostic	Faithful	NLL
Risk / actual risk25 benefit	-0.021 +/- 0.049	-0.052 +/- 0.013
Dense/actual risk25 sign disagreement (%)	30.201 +/- 0.587	31.303 +/- 0.452
Base strictly best in budgeted menu (%)	35.915 +/- 11.963	35.802 +/- 6.257

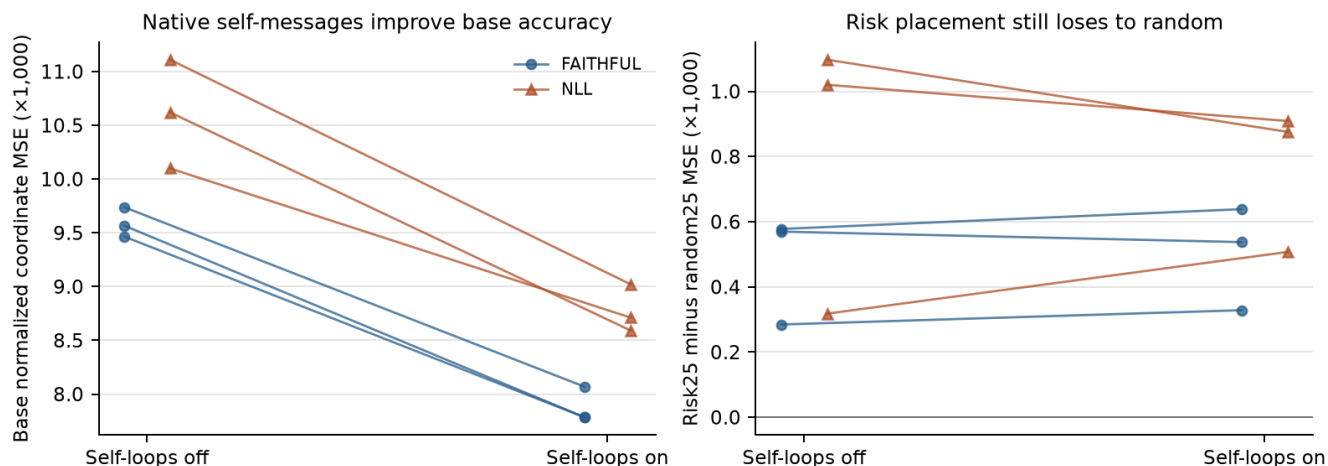
Dense benefit is an imperfect proxy for the sparse action, but replacing it with actual-action benefit does not recover strong risk ranking. Weak rank correlation does not exclude nonmonotone structure: the highest risk quartile has positive mean actual risk-action benefit for both objectives, with one negative NLL seed. Random allocation yields still greater benefit there in all six models.

The hindsight menu selects one complete whole-frame prediction from base, random25, speed25 and risk25. It uses the target and is not deployable. Base being best on roughly 36% of frames motivates learning when to add edges; a positive residual scale alone supplies no calibrated signed-gain threshold.

The reviewers identified an additional confound: full models trained on capped base graphs with self-loops and no expansion, while locked evaluation removed self-loops and added unseen annulus edges. The separate completed six-model, 16-case graph-convention screen is reported next. Native-graph autonomous controls are running; paired graph-exposure continuation is prepared but has not started.

14. Self-messages improve accuracy, but not risk placement

All six unchanged 100k checkpoints completed 425 observed histories each: 128 validation and 297 test, with 16 graph/policy cases per history. There were zero failed frames. This follow-up was frozen after inspection of the original study and remains exploratory.



Each line joins one training seed across self-loop settings. Right: positive differences favor random25 over previous-observed-base risk25. Risk scoring also changes with the loop setting; this is not a pure graph perturbation with fixed risk-selected pairs.

Paired diagnostic	Faithful	NLL
Validation base MSE reduction (%)	18.13 +/- 1.92	17.01 +/- 2.43
Validation risk-random MSE, loops on (x1,000)	0.878 +/- 0.203	1.581 +/- 0.525
Test base MSE reduction (%)	17.85 +/- 0.75	17.21 +/- 3.01
Test risk-random MSE, loops on (x1,000)	0.502 +/- 0.158	0.764 +/- 0.223

Means and sample SD are across three paired seed means, with equal-trajectory weighting. Relative reductions are computed within seed before averaging. Risk loses to random in every seed on both splits; dense is worse than base throughout. Current-base risk also does not beat random. All policies, loop/cap conditions and paired interactions are retained in the manuscript appendix and public results.

Cap128 never binds on the 425 observed geometries. Every native graph/feature parity gate passes; maximum prediction discrepancy is 5.96e-8. The earlier geometric inspection already found cap inactivity on the 297 test states. Bridge random allocation uses a new paired draw, so it is not an exact replay of the original random results. These checks do not establish cap inactivity during autonomous collapse.

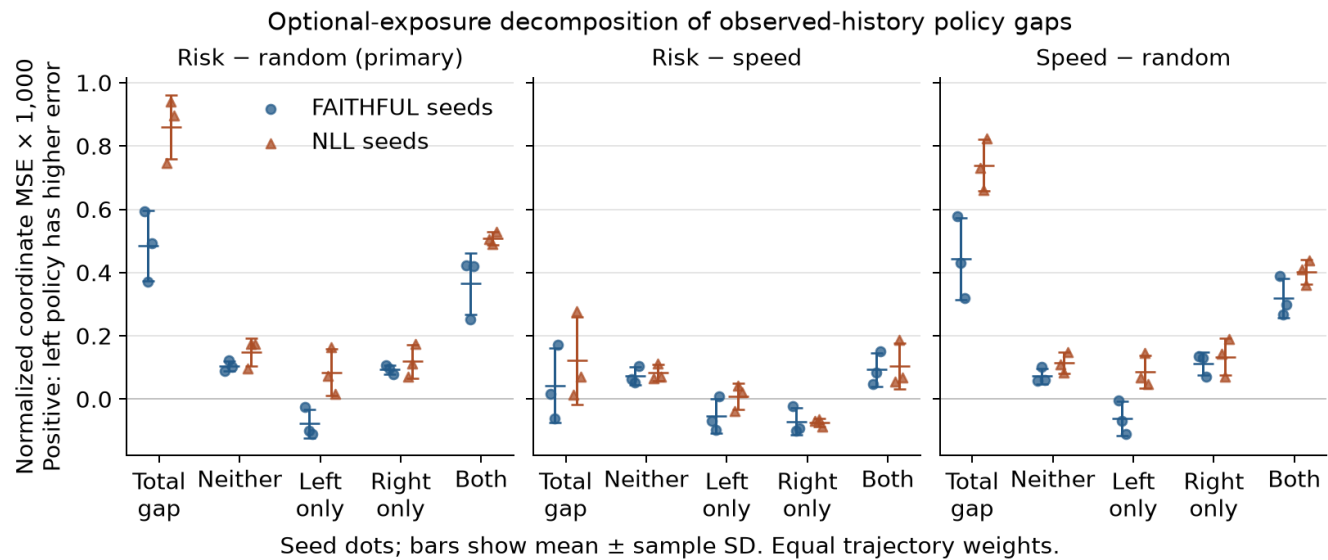
Interpretation: removing trained self-messages materially affected absolute accuracy, but did not explain the risk-random ordering. Lack of expanded-graph training remains unresolved. The running native autonomous controls and prepared equal-update graph-exposure continuations address feedback and training support; no results from either are claimed here.

Strict analysis passed 851,400 coverage/arithmetic/integrity checks; an independent raw-record aggregation passed 5,777 checks. All 2.24 GB of raw bridge evidence remain local and hashed; public compact artifacts preserve every case/seed summary and identify omitted arrays.

The six model processes took 693.156 seconds (11.55 minutes), including loading, inference, diagnostics and writing. Separate audits and document work are excluded. Fixed case order and shared work prevent interpreting this as policy latency.

15. Where the risk-random error gap appears

This separate exploratory analysis uses all 1,782 original observed-history frames and saved predictions. For each policy pair, it partitions particles by optional-edge exposure under neither action, the left action only, the right action only, or both. The four unconditional contributions sum to the whole error gap.



Positive values mean higher error for the left policy. Symbols show all six model seed means; bars show objective means and sample SDs across three seeds. Contributions average particles within frame, frames within trajectory, then trajectories within seed. These retain the original no-self-loop evaluation convention.

Risk minus random, MSE x1,000	Faithful	NLL
Whole error gap	0.486 +/- 0.111	0.860 +/- 0.102
Neither receives optional edges	0.105 +/- 0.016	0.148 +/- 0.044
Risk only	-0.077 +/- 0.046	0.084 +/- 0.073
Random only	0.094 +/- 0.014	0.119 +/- 0.053
Both	0.365 +/- 0.097	0.508 +/- 0.020

The both-covered group contributes the largest positive error difference in every seed. Risk-only particles favor risk in all three faithful models, but favor random in all three NLL models. The speed-random comparison also has a positive both-covered contribution in every seed. These statements describe the recorded partitions; they do not establish a causal effect of concentration.

Whole-frame difference, x1,000	Faithful	NLL
Squared change per coordinate	3.696 +/- 0.505	3.798 +/- 0.164
Alignment with base residual	3.210 +/- 0.607	2.938 +/- 0.073

All terms use normalized acceleration coordinates. Error difference equals the per-coordinate squared-change cost difference minus the residual-alignment difference. Risk induces more residual-aligned prediction changes, but their additional squared magnitude is larger in every seed. Exposure is defined by the selected action, and ten message-passing blocks transmit effects beyond directly incident optional edges; neither means unexposed to direct additions, not unaffected.

The CPU postprocess took 42.016 seconds with no new inference. An independent all-array audit took 34.107 seconds and passed 67,052,267 checks across all frames, with maximum arithmetic discrepancy 1.78e-15. All group/subgroup signs, empty-group nulls, fixed-weight conditional ratios and input identities remain in the saved evidence.

16. Native-convention autonomous follow-up

October 6 addendum. The first 16 pages retain their earlier dated snapshot. This addendum supersedes their statements that native-convention rollout controls are still running.

The same six fixed 100k WaterDrop models were evaluated on official test sources 3-29, under five policies, for 995 forecasts: 810 outcomes. All 405 faithful outcomes completed; seven NLL seed-2 outcomes hit coordinate guards.

This follow-up preserves the native directed, capped base prefix and self-edge candidates, then appends uncapped optional annulus messages. It changes graph conventions jointly. The test population was previously inspected; this is neither an isolated causal self-loop estimate nor pristine independent confirmation. Earlier evidence remains separate.

Full-rollout coordinate MSE by training seed. Each value equally averages 27 trajectory means. Parentheses count failures out of 27; all other cells have zero failures.

Policy	Faithful Seed 0	Faithful Seed 1	Faithful Seed 2	NLL Seed 0	NLL Seed 1	NLL Seed 2
Base	0.014756	0.010450	0.035972	0.025363	0.010036	undefined (1)
Dense	0.023553	0.022720	0.067671	0.044155	0.012284	undefined (1)
Random	0.016851	0.011730	0.044116	0.028950	0.008695	undefined (2)
Speed	0.017626	0.012091	0.036325	0.033088	0.009283	undefined (2)
Cached risk	0.018106	0.011947	0.043985	0.033657	0.009282	undefined (1)

The seven failures occurred at forecasts 861-969. **All five three-seed NLL full-horizon means are undefined.** No failure was retried; complete seeds are shown without replacing the full cohort by a survivor-only average.

Paired faithful effects across all three seeds

Comparator	Cached risk minus comparator: mean +/- sample SD
Base	+0.00428683 +/- 0.00335776
Random	+0.00044705 +/- 0.00072071
Speed	+0.00266514 +/- 0.00433674
Dense	-0.01330204 +/- 0.00937878

Positive differences mean higher error for cached risk. These means and sample SDs are descriptive, not significance claims. Cached risk loses to base in all three seeds; its random and speed contrasts change sign. Dense loses to base in every seed. Matching the trained base convention does not establish a consistent allocation advantage.

Boundary diagnostics, runtime and provenance

Faithful native-convention rollouts: mean +/- sample SD over three training seeds. Outside fraction averages particles and forecast steps within trajectory, then trajectories equally within seed. Excursion averages each trajectory's maximum coordinate excursion.

Policy	Outside (%)	Mean trajectory-max excursion
Base	8.54 +/- 4.14	0.173 +/- 0.091
Dense	11.16 +/- 5.01	0.421 +/- 0.264
Random	9.25 +/- 4.68	0.238 +/- 0.128
Speed	8.50 +/- 4.00	0.256 +/- 0.154
Cached risk	8.69 +/- 4.19	0.251 +/- 0.136
Truth	2.68 +/- 0.00	0.00782 +/- 0.00000

The outside threshold is 1e-6 relative to the declared domain. These are geometric diagnostics, not certified physical failure labels. The repeated truth cohort has zero between-seed SD by construction. Model excursions exceed truth, and numerical completion does not establish physical validity. The native cap never binds in completed faithful trajectories.

Measured clocks and their limits

Saved runtime counters include graph construction, network calls and diagnostics. The fixed policy order and policy-dependent rollout geometries prevent a fair optimized speedup claim. Graph, forward and parity counters remain available in the exact summary.

The NLL seed-2 process was suspended and resumed at the author's request. Its raw clocks are preserved as interrupted elapsed time and excluded from uninterrupted runtime claims. No guessed pause duration is subtracted, and convenient trajectories are not substituted.

Audited evidence and reproducibility

The strict source, trace and arithmetic audit passed 19,521,125 checks. A complementary, independently implemented scalar and paired audit passed 1,724,434 checks. These counts certify saved-artifact consistency; they do not establish scientific validity, calibration or independent statistical confirmation.

The compressed public summary retains all seeds, policies, failed prefixes, boundary results and comparisons to the original convention. Its identities link the retained source, configuration, checkpoints and raw records. The independent audit binds the same summary. Unstored intermediate model outputs cannot be independently regenerated without inference.

Source folder: **research/results/native_rollout_completion_20261006/**

Summary: **native_graph_rollouts.json.gz**

Inflated SHA256: 23ecd bba1c906043530519c9ae74bcc1
67fda074ad4c01470838271fc2984e1f

Audit: **independent_scalar_audit.json.gz**

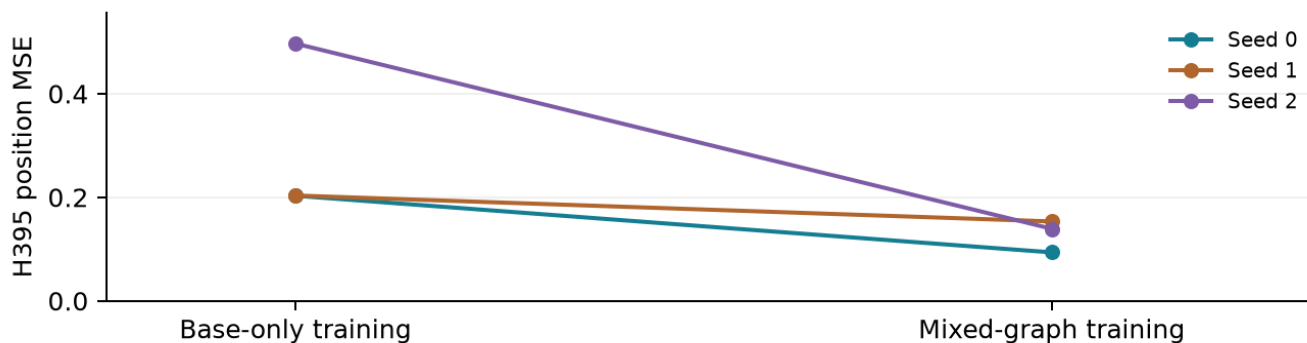
Inflated SHA256: de17d9016d76575534dabaabf27885f0
79c3117d756698bd4443a968fcc005fc

Goop2D research is in progress; 3D work remains preparation only. Neither contributes efficacy evidence here. Author verification and submission checks remain outstanding. This report does not certify conference readiness.

17. Graph exposure: fresh Goop training

October 6 scientific update. The preceding 18 pages retain their earlier dated snapshot. This addendum supersedes their in-progress descriptions of Goop2D100k and WaterDrop110k.

Six faithful models train from initialization to 100k updates on all 1,000 official training trajectories. Paired arms share initialization, sampled frames and noise. Mixed training expands half of examples with a uniform quarter of optional annulus pairs; native base messages remain intact.



Fixed-policy improvement. Random25 full-rollout MSE improves in every paired seed. Lines connect the same seed; each value averages 30 trajectory means. The graph-exposure intervention is separate from choosing where to place extra messages.

Inference policy	Base-only: mean +/- SD	Mixed: mean +/- SD	Mix minus base: mean +/- SD
Base	0.247 +/- 0.149	undefined	undefined
Dense	0.496 +/- 0.237	undefined	undefined
Random25	0.301 +/- 0.169	0.129 +/- 0.031	-0.172 +/- 0.163
Speed25	0.272 +/- 0.135	0.112 +/- 0.027	-0.160 +/- 0.137
Relative-velocity RMS25	0.333 +/- 0.209	0.125 +/- 0.032	-0.208 +/- 0.204
Cached risk25	0.310 +/- 0.157	undefined	undefined

Coverage and interpretation

All 1,080 planned H395 outcomes are retained: 1,077 complete and three candidate-pair guards, one each for mixed seed 2 base, dense and cached risk. Their full three-seed means and the risk-versus-random training interaction remain undefined. No survivor mean replaces them.

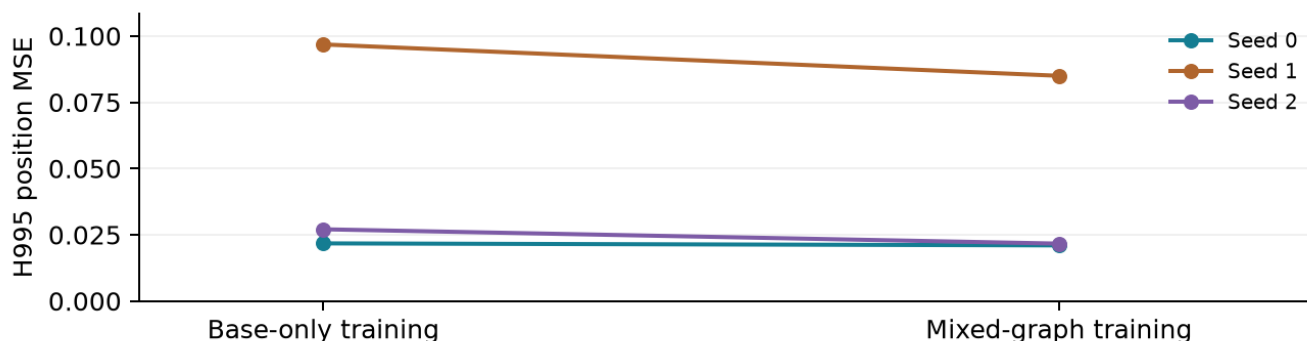
On matched observed test histories, risk still loses to random in every mixed-model seed. Mixed speed and relative-velocity RMS beat random in every full-rollout seed. Thus exposure helps this simulator, while residual ranking is not established as a superior placement rule.

Mixed random, speed and RMS retain 22.16%, 19.55% and 20.53% of particles outside the metadata box by more than $1e-6$, versus 0.0632% for truth. Box excursions do not test conservation. Shared-host, policy-dependent rollout timing does not establish isolated speedup.

Source: `goop2d_graph_exposure_100k_20261006/paired_scalar_summary.json.gz` and its admitted saved-array audits. Checks cover saved positions and scalar series; unsaved trajectories and source/model execution are not replayed. Three-seed means and sample SDs are descriptive.

18. Graph exposure: WaterDrop continuation

Three fixed faithful 100k parents each receive paired 10k base-only and mixed-graph continuations, inheriting optimizer state and matched frame/noise schedules. The endpoint is 110k; this is conditional continuation evidence, not fresh 110k training.



Fixed-policy improvement. Random25 improves in every paired seed, as do dense, speed and cached risk. Each seed equally averages 27 trajectory means. Different material scales and training budgets prevent an absolute-error comparison with Goop.

Inference policy	Base-only: mean +/- SD	Mixed: mean +/- SD	Mix minus base: mean +/- SD
Base	0.0444 +/- 0.0380	0.0390 +/- 0.0302	-0.0054 +/- 0.0078
Dense	0.0548 +/- 0.0361	0.0432 +/- 0.0333	-0.0115 +/- 0.0040
Random25	0.0485 +/- 0.0419	0.0425 +/- 0.0367	-0.0060 +/- 0.0056
Speed25	0.0401 +/- 0.0303	0.0351 +/- 0.0247	-0.0050 +/- 0.0057
Cached risk25	0.0418 +/- 0.0303	0.0387 +/- 0.0294	-0.0032 +/- 0.0009

Placement and physical behavior

All 810 H995 outcomes complete with zero scientific failures. All 2,550 observed histories and 12,750 policy slots are retained. Observed-test risk changes from worse than random to better in every seed after exposure. Validation does not share a universal reversal: mixed risk remains worse than random on average.

The autonomous risk-minus-random training interaction is $+0.00284 \pm 0.00466$, with mixed seed signs. Observed-state scores are computed on a preceding observed base history; autonomous cached scores follow the policy's own graph. An observed-state gain therefore does not establish a consistent autonomous placement benefit.

Mixed random, speed and cached-risk rollouts still place more than 16% of particles outside the metadata box, versus 2.68% for truth. Recorded times and edge counts increase for every policy after exposure. These fixed-order, geometry-dependent costs remain descriptive. Numerical completion does not certify physical validity.

Scientific takeaway

Across these two exploratory studies, training on expanded graphs improves full rollouts under an unchanged random policy in every paired seed. Useful placement is a separate question, with adverse or mixed risk results. The evidence supports graph exposure as a constructive intervention; it does not establish universal adaptive-allocation or efficiency superiority.

Source: [waterdrop_continuation_110k_20261006/summary/result.json](#), its 91 metric-detail files and independent audit. All earlier fixed-study evidence remains unchanged. This addendum includes no interim Sand or Goop 3D accuracy claims.

19. Graph exposure: Sand

Completed 100k study | six faithful models, three paired seeds | all 1,080 full-H314 rollouts complete

Base-only and mixed-graph models train from initialization on all 1,000 training trajectories. Paired arms share initialization, sampled frames and noise. Mix expands half of training examples with a random quarter of optional annulus pairs; the native capped base graph remains intact.

Inference policy	Base-only mean +/- SD	Mixed graph mean +/- SD	Mix minus base mean +/- SD [seed signs]
Base	0.06785 +/- 0.06378	0.06781 +/- 0.06218	-0.00004 +/- 0.00263 [-/-+]
Dense	0.12934 +/- 0.14479	0.12602 +/- 0.14751	-0.00332 +/- 0.00296 [-/-]
Random25	0.08298 +/- 0.08643	0.07240 +/- 0.06802	-0.01058 +/- 0.01861 [-/-+]
Speed25	0.08085 +/- 0.07898	0.09071 +/- 0.09794	+0.00986 +/- 0.01898 [-/+ +]
Cached risk25	0.09714 +/- 0.10755	0.10631 +/- 0.11951	+0.00917 +/- 0.01201 [+/+ +]
Relative-velocity RMS25	0.08782 +/- 0.09293	0.08507 +/- 0.08888	-0.00275 +/- 0.00498 [-/-+]

Full-horizon position MSE; each seed equally averages all 30 trajectory means. Means and sample SDs describe three seeds. Signs are ordered seeds 0/1/2; a minus favors mixed training. All six policies and all required outcomes are retained.

Useful training effect, conditional expansion. Exposure lowers observed-test error under all five expanded policies in every seed; dense full-rollout error also improves in every seed. Yet native base beats both random25 and dense in every seed of both arms. Learning to use an expanded graph does not imply that expansion outperforms the native graph.

Residual ranking does not transfer through feedback

On matched observed test histories, the risk-minus-random deficit narrows from +4.76 +/- 3.80 to +1.87 +/- 1.04 (position-coordinate MSE x 10⁻⁹), but risk remains worse in every seed of both arms. In autonomous rollouts, exposure worsens cached-risk error in every seed and widens its risk-minus-random gap by +0.01975 +/- 0.03062, also adverse in every seed. Observed risk uses the preceding observed base graph; autonomous risk caches its own selected-graph scores.

Completion still leaves physical and cost limitations

Mixed random25 places 10.47% of particles outside the metadata box by more than 1e-6 (truth: 0.0618%), with mean trajectory-maximum excursion 0.3244 (truth: 0.0007954). Across all policies, mean outside fractions fall after exposure, but maximum excursions increase in every paired seed. These are geometric diagnostics, not conservation tests.

Mean committed-call time increases for all six policies; random25 changes from 14.06 +/- 0.90 to 14.70 +/- 1.66 seconds/case. Calls include native parity. Fixed policy order, shared hosts and evolving geometries prevent an isolated speedup claim.

Coverage and scope

All 3,648 mixed-stage cells complete: 1,080 autonomous outcomes, 900 observed validation histories, 900 observed test histories and 768 clean-validation frames. None failed or remained uncompleted. Sand is an exploratory, distinct native-CUDA lineage; earlier strict CPU/CUDA gradient checks failed, so CPU equivalence and a controlled material effect are not claimed.

Source: completed Sand100k paired summary and audited saved-array/scalar evidence; manuscript appendix, "Sand: graph exposure and placement." The preceding 20 report pages preserve their dated snapshots. This page contains no Goop3D numerical claim.